

DEEN DAYAL UPADHYAYA GORAKHPUR UNIVERSITY

Faculty of Engineering & Technology - Institute of Engineering & Technology

NAAC Accredited Grade "A++" | Established 1957 | Gorakhpur, UP, India

OFFICIAL CURRICULUM & DEGREE STRUCTURE - B.TECH INFORMATION TECHNOLOGY

Four-Year Undergraduate Degree Program | Choice Based Credit System (CBCS) & NEP 2020

Effective Academic Sessions: 2024-25, 2025-26 | Department of Information Technology

1. PROGRAM EDUCATIONAL OBJECTIVES (PEOs) & VISION

The B.Tech in Information Technology program aims to cultivate industry-ready software engineers, systems architects, cybersecurity specialists, and full-stack cloud developers equipped with deep foundational knowledge and modern engineering competencies.

2. TOTAL CREDIT DISTRIBUTION & DEGREE REQUIREMENTS

- Basic Sciences & Humanities (BSH): 28 Credits
- Engineering Science Courses (ESC): 24 Credits
- Professional Core Courses (PCC - IT): 68 Credits
- Professional Elective Courses (PEC): 18 Credits
- Open Elective Courses (OEC): 12 Credits
- Project Work, Seminar & Industrial Internship: 16 Credits
- Mandatory Non-Credit Audit Courses (Constitution of India, Essence of Indian Knowledge): Pass

Total Degree Completion Requirement: 166 Credits

3. FIRST YEAR CURRICULUM OVERVIEW (SEMESTER I & II)

Common foundational engineering courses covering Engineering Mathematics I & II, Engineering Physics, Engineering Chemistry, Programming for Problem Solving (C Language), Basic Electrical Engineering, Basic Electronics Engineering, Engineering Graphics & CAD, and Professional Communication.

B.TECH INFORMATION TECHNOLOGY - SEMESTER-WISE COURSE STRUCTURE

Curriculum Breakdown: Semesters 3 to 6 Core Specializations

SEMESTER 3 (Credits: 22)

1. BIT-301: Discrete Mathematical Structures & Graph Theory (3-1-0) - 4 Credits
2. BIT-302: Data Structures using C/C++ & Algorithms (3-1-0) - 4 Credits
3. BIT-303: Computer Organization & Architecture (3-0-0) - 3 Credits
4. BIT-304: Object Oriented Programming in Java & Modern C++ (3-0-0) - 3 Credits
5. BEC-301: Digital Electronics & Logic Design (3-0-0) - 3 Credits
6. BHS-301: Technical Communication & Universal Human Values (2-0-0) - 2 Credits
7. BIT-351: Data Structures & OOP Laboratory (0-0-3) - 1.5 Credits
8. BIT-352: Digital Logic & Hardware Simulation Lab (0-0-2) - 1.5 Credits

SEMESTER 4 (Credits: 23)

1. BIT-401: Operating Systems & Kernel Architecture (3-1-0) - 4 Credits
2. BIT-402: Theory of Automata & Formal Languages (3-1-0) - 4 Credits
3. BIT-403: Software Engineering & Agile Methodologies (3-0-0) - 3 Credits
4. BIT-404: Database Management Systems & SQL Engineering (3-1-0) - 4 Credits
5. BIT-405: Web Technologies & Scripting Frameworks (3-0-0) - 3 Credits
6. BIT-451: Operating Systems & Linux System Calls Lab (0-0-3) - 1.5 Credits
7. BIT-452: DBMS & Full-Stack Web Development Lab (0-0-3) - 1.5 Credits

SEMESTER 5 (Credits: 22)

1. BIT-501: Design & Analysis of Computer Algorithms (3-1-0) - 4 Credits
2. BIT-502: Computer Networks & Protocol Engineering (3-1-0) - 4 Credits
3. BIT-503: Information Security & Network Defense (3-0-0) - 3 Credits
4. BIT-504: Python for Data Science & AI Systems (3-0-0) - 3 Credits
5. Professional Elective I: Cloud Computing / Microservices / DevOps (3-0-0) - 3 Credits
6. BIT-551: Computer Networks & Socket Programming Lab (0-0-3) - 1.5 Credits
7. BIT-552: Algorithms & Competitive Programming Lab (0-0-3) - 1.5 Credits

SEMESTER 6 (Credits: 22)

1. BIT-601: Cloud Computing Infrastructure & Virtualization (3-1-0) - 4 Credits
2. BIT-602: Cryptography & Network Security (3-1-0) - 4 Credits
3. BIT-603: Mobile Application Development (Flutter/Android) (3-0-0) - 3 Credits
4. Professional Elective II: Big Data Analytics / Deep Learning (3-0-0) - 3 Credits
5. Open Elective I: Engineering Economics & Management (3-0-0) - 3 Credits
6. BIT-651: Cloud Deployment & Mobile App Lab (0-0-3) - 1.5 Credits
7. BIT-652: Mini Project & Industrial Summer Training Evaluation - 2 Credits

FINAL YEAR CURRICULUM & DEGREE ORDINANCES - B.TECH IT

Semesters 7 & 8: Industry Specialization, Capstone & Assessment Policy

SEMESTER 7 (Credits: 20)

1. BIT-701: Artificial Intelligence & Machine Learning (3-1-0) - 4 Credits
2. BIT-702: Internet of Things (IoT) & Smart Embedded Systems (3-0-0) - 3 Credits
3. Professional Elective III: Cyber Forensics & Malware Analysis (3-0-0) - 3 Credits
4. Open Elective II: Entrepreneurship & Intellectual Property Rights (3-0-0) - 3 Credits
5. BIT-751: AI & IoT Laboratory (0-0-3) - 1.5 Credits
6. BIT-752: Industrial Internship / Summer Training (0-0-4) - 2 Credits
7. BIT-753: Capstone Major Project - Phase I (Synopsis & Design) - 3.5 Credits

SEMESTER 8 (Credits: 18)

1. BIT-801: Distributed Computing & Blockchain Technologies (3-1-0) - 4 Credits
2. Professional Elective IV: Natural Language Processing / Full-Stack Next.js (3-0-0) - 3 Credits
3. Professional Elective V: Site Reliability Engineering & DevOps (3-0-0) - 3 Credits
4. BIT-851: Capstone Major Project - Phase II (Implementation & Defense) - 6 Credits
5. BIT-852: Comprehensive Technical Viva-Voce & Grand Evaluation - 2 Credits

EXAMINATION & CONTINUOUS EVALUATION SCHEME

Theory Courses: 100 Marks (30 Marks Internal Assessment + 70 Marks University End-Term Exam).

Practical Labs: 100 Marks (50 Marks Continuous Lab Assessment + 50 Marks External Lab Viva).

Minimum Passing Criteria: 40% in each individual theory & lab component, 50% cumulative aggregate.

OFFICIAL SIGNATURES & VERIFICATION STAMP:

Prof. Dean, Faculty of Engineering & Technology
Deen Dayal Upadhyaya Gorakhpur University, UP

Controller of Examinations
DDU Gorakhpur University